CS-4085 MLOps (Sections A, B)

Monday, February 23rd, 2026

Sessional 1
Total Time: 1 Hour
Total Marks: 43

Course Instructor(s)

Hammad Majeed

Student Name	Roll No	Section	Signature

Question 1 ..... (30 Marks)

Fill the right option on the answer sheet provided on page 4

(1) **(2 Marks)** You are building a Docker image for a scikit-learn model. You want to ensure that if your `train.py` code changes, Docker does NOT have to re-download all Python packages. Which Dockerfile sequence is most efficient?

A. `COPY . /app` followed by `RUN pip install -r requirements.txt`

B. `RUN pip install -r requirements.txt` followed by `COPY . /app`

C. `COPY requirements.txt .`, `RUN pip install -r requirements.txt`, then `COPY . /app`

D. `RUN pip install scikit-learn` at the very end of the file

(2) **(2 Marks)** Your GitHub Actions workflow needs to push a Docker image to GitHub Container Registry (GHCR). To do this securely, where should the `CR_PAT` (Personal Access Token) be stored?

A. Hardcoded in the `main.yml` workflow file

B. As a plain-text environment variable in the repository settings

C. In GitHub Secrets and referenced via `${{ secrets.CR_PAT }}`

D. In a `.env` file committed to the Git repository

(3) **(2 Marks)** In a Docker Compose setup, your `fastapi-app` service cannot connect to your `postgres-db` service using the hostname `db`. What is the most likely cause?

A. The services are not on the same Docker network.

B. You forgot to map port 5432 to the host machine.

C. Docker Compose does not support service name resolution.

D. The `postgres-db` service is using a different image tag.

(4) **(2 Marks)** You have a CI pipeline with two jobs: `lint` and `test`. You want to save compute time by ensuring `test` only runs if `lint` passes. Which YAML key is required?

A. `needs: lint`

B. `if: success()`

C. `runs-on: ubuntu-latest`

D. `strategy: matrix`

(5) **(2 Marks)** A Docker container exits with code 137 while processing a large 10GB dataset. What is the most appropriate fix?

A. Fix the syntax error in the Python script.

B. Increase the memory limit assigned to the container.

C. Change the Docker base image to a lighter version.

D. Re-build the image without cache.

(6) **(2 Marks)** A Docker Compose file defines a service `api` with `ports: - "8080:5000"`. A second service named `worker` needs to send a request to the `api` service. What is the correct URL for the worker to use?

A. `http://localhost:8080`

B. `http://api:8080`

C. `http://api:5000`

D. `http://localhost:5000`

(7) **(2 Marks)** You are using GitHub Environments to manage deployments. You have a secret named `DATABASE_URL` defined in the "Production" environment and a different `DATABASE_URL` in the "Staging" environment. How does GitHub Actions know which one to use?

A. It automatically picks based on the branch name.

B. You must specify `environment: Production` or `environment: Staging` in the job definition.

C. You must use an `if` statement to check the branch and then manually map the secret.

D. It is not possible to have two secrets with the same name in the same repository.

(8) **(2 Marks)** Which of the following `on:` triggers in GitHub Actions is most effective for a repository that contains both documentation and ML code, to avoid wasting CI minutes when only the `README.md` is updated?

A. `on: push: branches: [main]`

B. `on: push: paths-ignore: ['**.md']`

C. `on: pull_request: types: [opened]`

D. `on: workflow_dispatch`

(9) **(2 Marks)** In a Multi-stage Dockerfile, what is the primary purpose of using the `FROM ... AS build` and `FROM ... AS runtime` pattern?

A. To allow the container to run two processes at the same time.

B. To speed up the initial pull of the base image.

C. To create a final production image that contains only the app and its dependencies, excluding build tools and source caches.

D. To bypass the need for Docker Compose.

(10) **(2 Marks)** When deploying a model to production via CD, why is it considered a "Bad Practice" to use the `:latest` tag for your Docker images?

A. The `:latest` tag makes the image file size larger.
B. It breaks traceability; you cannot easily roll back to a specific known-good version if the new "latest" fails.
C. Docker Hub charges extra for using the `:latest` tag.
D. The `:latest` tag prevents the container from starting automatically.

(11) **(2 Marks)** You have a `docker-compose.yml` where a service `db` uses a volume: `- ./data:/var/lib/postgresql/data`. What happens to the data if you run `docker-compose down` and then `docker-compose up`?

A. The data is deleted and the database starts empty.
B. The data is preserved because it is stored on the host machine's `./data` directory.
C. The data is only preserved if the container is "paused" instead of "downed."
D. Docker Compose will throw an error because the directory already exists.

(12) **(2 Marks)** In a GitHub Actions workflow, you want to run your unit tests against Python 3.8, 3.9, and 3.10 simultaneously. Which feature should you use?

A. `steps: matrix`
B. `jobs: parallel`
C. `strategy: matrix`
D. `runs-on: [3.8, 3.9, 3.10]`

(13) **(2 Marks)** Your CD pipeline uses SSH to deploy an image to a remote server. The workflow fails at the "Login to Docker" step because it cannot find the credentials. What is the most likely missing step?

A. You didn't install Docker on the GitHub Runner.
B. You didn't use `docker/login-action` with your Registry secrets.
C. The remote server is offline.
D. You didn't expose the ports in the Dockerfile.

(14) **(2 Marks)** What is the effect of the following line in a GitHub Actions job?
`continue-on-error: true`

A. The workflow will automatically retry the step until it passes.
B. The job will be marked as "Passed" even if this specific step fails, allowing subsequent steps to run.
C. The workflow will pause and wait for manual intervention.
D. It prevents the job from ever failing, regardless of any errors in any step.

(15) **(2 Marks)** A pytest suite passes locally but fails in CI due to missing environment variables. Which solution is MOST robust?

A. Hardcode defaults in test files
B. Skip tests when variables are missing
C. Inject variables via CI configuration
D. Store variables in Git

NAME :
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Roll No

0	○	○	○	○	○	○	○
1	○	○	○	○	○	○	○
2	○	○	○	○	○	○	○
3	○	○	○	○	○	○	○
4	○	○	○	○	○	○	○
5	○	○	○	○	○	○	○
6	○	○	○	○	○	○	○
7	○	○	○	○	○	○	○
8	○	○	○	○	○	○	○
9	○	○	○	○	○	○	○

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Section1

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○

	A	B	C	D
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○

	A	B	C	D
11	○	○	○	○
12	○	○	○	○
13	○	○	○	○
14	○	○	○	○
15	○	○	○	○

Question 2 ..... **(5 Marks)**

Task: Design a GitHub Actions workflow file (.github/workflows/deploy.yml) for an ML project that automates the following requirements:

- Trigger: Only on a push to the release branch.
- Job 1 (CI): Install dependencies and run pytest.
- Job 2 (CD): Depends on Job 1.
  - Builds a Docker image tagged with the commit SHA.
  - Logs into Docker Hub using secrets.
  - Pushes the image to the registry.

Structure your answer as a high-level YAML sequence or a detailed list of steps.

Solution:

```
name: CI-CD Pipeline

on:
 push:
  branches:
   - release

jobs:
ci:
 name: Run Tests
 runs-on: ubuntu-latest

 steps:
  - name: Checkout Code
    uses: actions/checkout@v3

  - name: Set up Python
    uses: actions/setup-python@v4
    with:
    python-version: '3.10'

  - name: Install Dependencies
    run: |
   pip install -r requirements.txt

  - name: Run Tests
    run: pytest

cd:
 name: Build and Push Docker Image
 runs-on: ubuntu-latest
```

```
    needs: ci

    steps:
     - name: Checkout Code
       uses: actions/checkout@v3

     - name: Log in to Docker Hub
       uses: docker/login-action@v2
       with:
      username: ${{ secrets.DOCKER_USERNAME }}
      password: ${{ secrets.DOCKER_PASSWORD }}

     - name: Build Docker Image
       run: |
      docker build -t ${{ secrets.DOCKER_USERNAME }}/ml-app:${{ github.sha }} .

     - name: Push Docker Image
       run: |
      docker push ${{ secrets.DOCKER_USERNAME }}/ml-app:${{ github.sha }}
```

Question 3 ..... **(8 Marks)**

Case Study: The Silent Database (Docker Compose Networking)
Scenario: You are deploying a three-tier ML application:

- web-app: A Flask frontend.
- model-api: A FastAPI service that serves predictions.
- db: A PostgreSQL database to log every prediction.

In your docker-compose.yml, the model-api tries to connect to the database using localhost:5432. The container logs show: ConnectionRefusedError: [Errno 111] Connection refused.

(1) **(2 Marks)** Within a Docker Compose network, what should the connection string for the database be instead of localhost?

Solution: $postgresql://user:password@db:5432/database_name$

(2) **(4 Marks)** You notice that model-api crashes because it tries to connect before the db is fully "Ready." How would you ensure that the db is up and healthy before the API starts connecting to the db?

Solution:

```
services:
 db:
  image: postgres:latest
  healthcheck:
```

```
      test: ["CMD-SHELL", "pg_isready -U postgres"]
      interval: 5s
      retries: 5

    model-api:
     build: .
     depends_on:
      db:
       condition: service_healthy
```

(3) **(2 Marks)** Your team wants to ensure the db is not accessible from the public internet, but the web-app is. How do you configure the ports section for each service to achieve this?

Solution:

```
services:
 web-app:
  ports:
   - "80:5000"   # exposed to public

 model-api:
  ports:
   - "8000:8000"  # optional (internal or limited exposure)

 db:
  # NO ports section → not exposed publicly
```